Sri Lanka Institute of Information Technology

Faculty of Computing

Integration

Ms.Nilushi Dias

Year 01 and Semester 01

Outline

1. Integration
2. Standard integrals
3. Integration of polynomial expressions
4. Functions of a linear function of x
5. Integration by partial fractions
6. Areas under curves
7. Integration as a summation

Integration

Constant of integration

- Integration is the reverse process of differentiation. For example:

$$\frac{d}{dx}(x^4) = 4x^3$$

- The integral of $4x^3$ is then written as:

$$\int 4x^3.dx$$

- Its value is, however:

$$\int 4x^3.dx = x^4 + C \text{ where } C \text{ is called the } \textit{constant of integration}$$

Standard integrals

- Just as with derivatives we can construct a table of standard integrals:

$f(x)$	$\int f(x).\mathrm{d}x$
x^n	$\frac{x^{n+1}}{n+1}+C$
1	$x+C$
a	$ax+C$
$\sin x$	$-\cos x+C$
$\cos x$	$\sin x+C$
$\sec^2 x$	$\tan x+C$
e^x	e^x+C
a^x	$a^x/\ln a+C$
$\frac{1}{x}$	$\ln x+C$

Integration of polynomial expressions

- Just as polynomials are differentiated term by term so they are integrated, also term by term. For example:

$$\int\left(4x^3+5x^2-2x+7\right).dx = x^4+\frac{5x^3}{3}-x^2+7x+C$$

Functions of a linear function of x

- To integrate

$$\int (ax+b)^n .dx$$

- we change the variable by letting u = ax + b so that du = a.dx. Substituting into the integral yield:

$$\begin{aligned}\int (ax+b)^n .dx &= \int u^n .du \\ &= \frac{u^{n+1}}{n+1} + C \\ &= \frac{(ax+b)^{n+1}}{n+1} + C\end{aligned}$$

Integration by partial fractions

To integrate $\int \frac{7x+8}{2x^2+11x+5}dx$ we note that $\frac{7x+8}{2x^2+11x+5} = \frac{3}{x+5} + \frac{1}{2x+1}$

so that:

$$\int \frac{7x+8}{2x^2+11x+5}dx = \int \frac{3}{x+5}dx + \int \frac{1}{2x+1}dx$$

These partial fractions are 'functions of a linear function of x'

based on the standard integral $\int \frac{1}{x}dx$ and so

$$\int \frac{7x+8}{2x^2+11x+5}dx = 3\ln(x+5) + \frac{1}{2}\ln(2x+1) + C$$

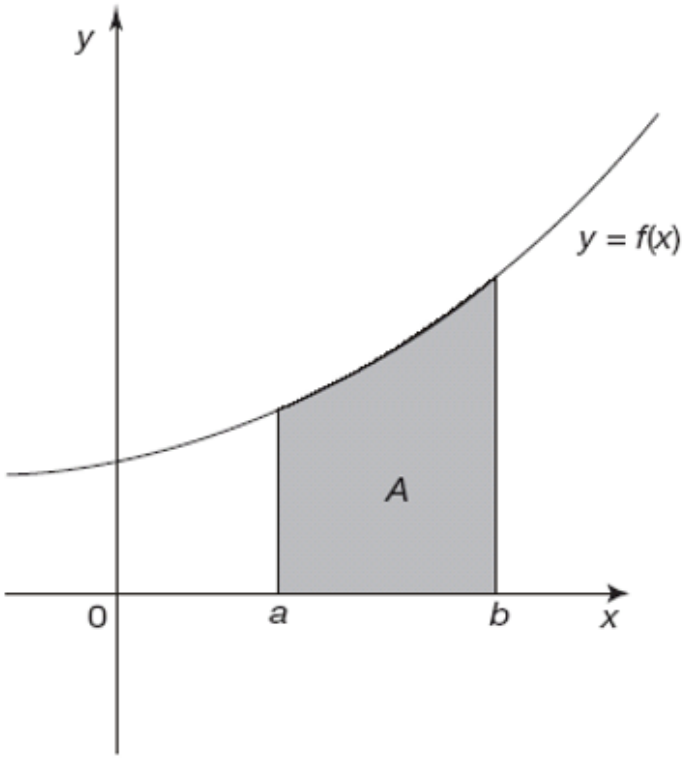
Areas under curves

- Area A, bounded by the curve y = f(x), the x-axis and the ordinates x = a and x = b, is given by:

$$A = \int_{x=a}^{b} f(x)dx$$
$$= \left[F(x)\right]_{x=a}^{b}$$
$$= F(b) - F(a)$$

Where,

$$F'(x) = f(x)$$



Thank You!